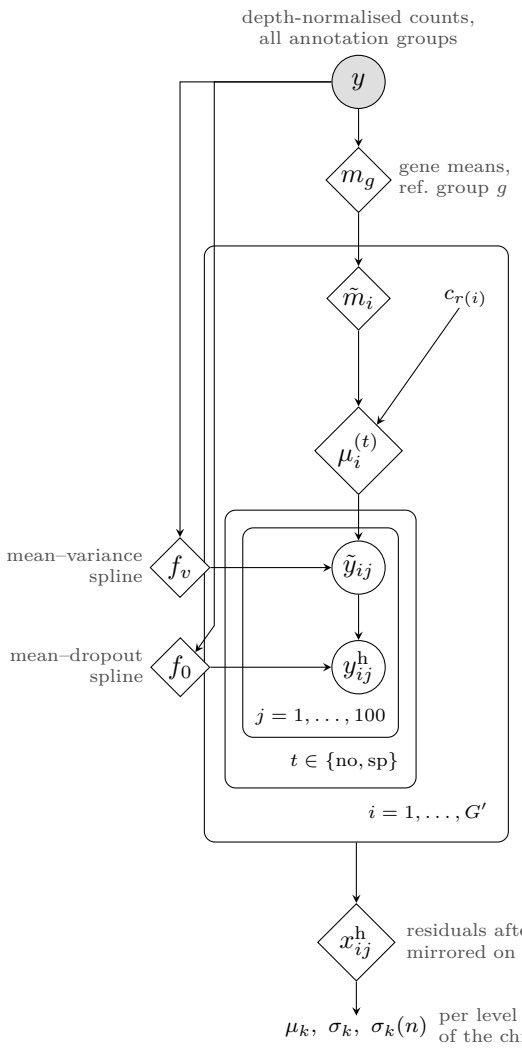
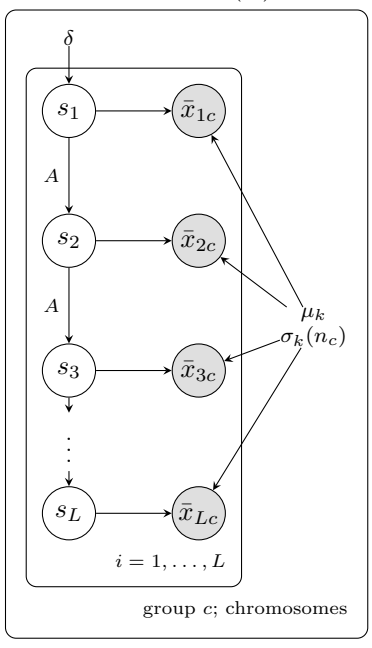
a h-spike simulator (i6)

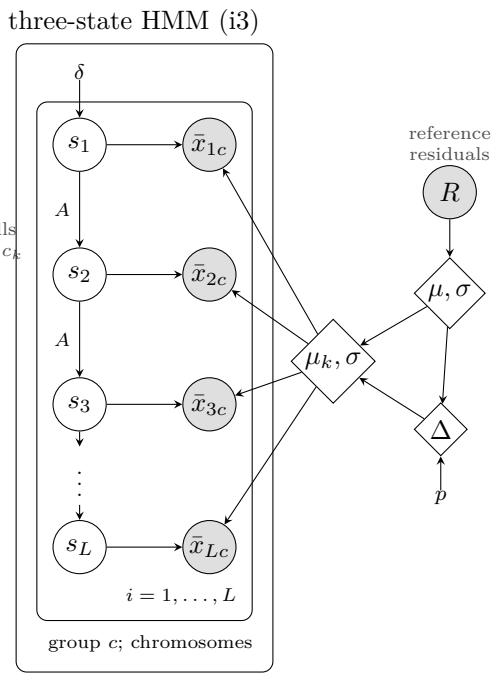


b six-state HMM (i6)



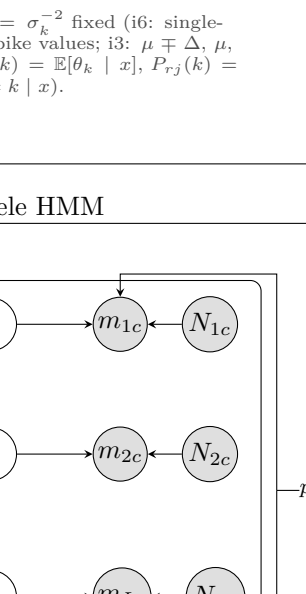
$K = 6$ states, $k = 3$ neutral; μ_k, $\sigma_k(n_c)$ from the h-spike (a); $\bar{x}_{ic}$: mean residual of the n_c cells of group c at gene i; $A_{kk} = 1 - 5t$, $A_{kl} = t$.

c three-state HMM (i3)



$K = 3$ states (loss, neutral, gain) with means $\mu - \Delta,\ \mu,\ \mu + \Delta$ and common σ, not scaled with n_c; $\Delta = \sigma\Phi^{-1}(1 - p)$ or the KS shift; $A_{kk} = 1 - 2t$.

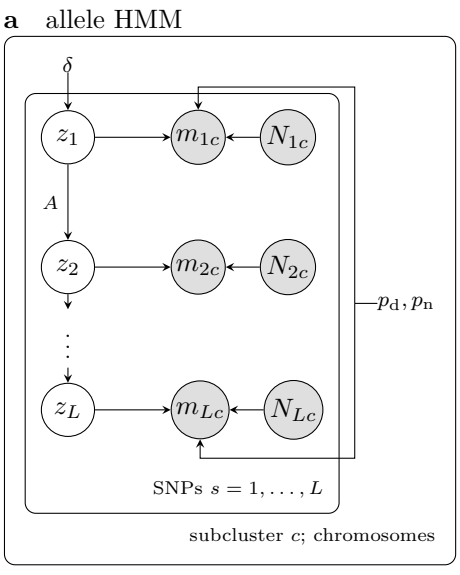
d Bayesian latent mixture (i6 and i3)



μ_k, $\tau_k = \sigma_k^{-2}$ fixed (i6: single-cell h-spike values; i3: $\mu \mp \Delta$, μ, σ). $P_r(k) = \mathbb{E}[\theta_k \mid x]$, $P_{rj}(k) = \Pr(\varepsilon_j = k \mid x)$.

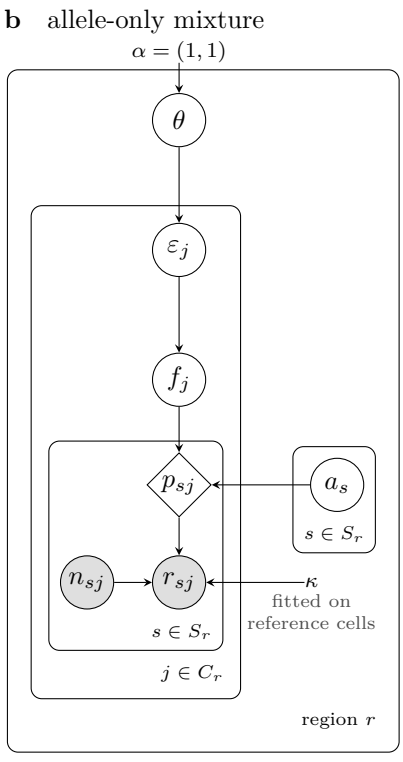
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a allele HMM



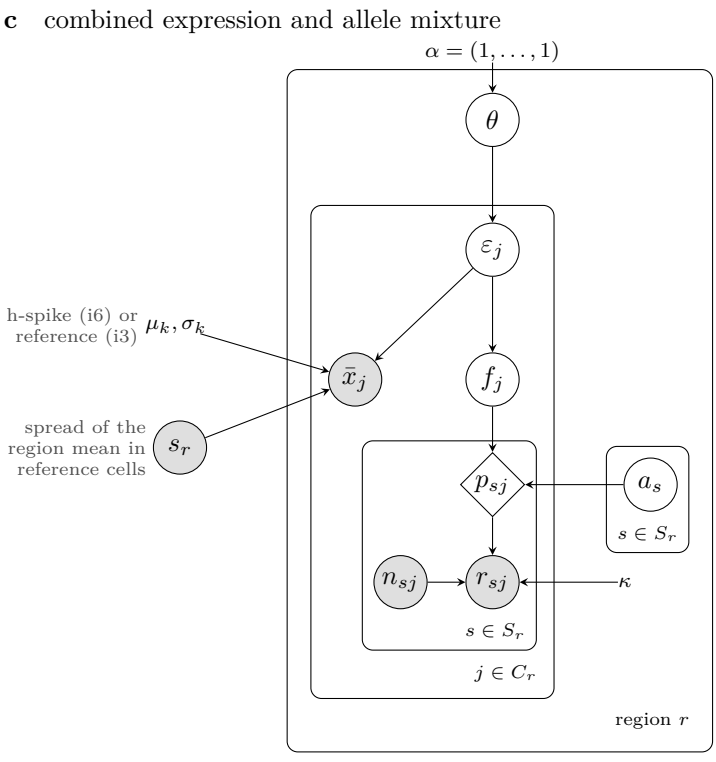
$z_s \in \{\text{imbalanced, balanced}\}$; of the N_{sc} cells of c covering SNP s, m_{sc} show the lesser allele: $m_{sc} \sim \mathrm{Bin}(N_{sc}, p_{z_s})$, $p_{\mathrm{d}} = 0.1$, $p_{\mathrm{n}} = 0.45$; $A_{kk} = 1 - t$; $\delta = (\frac{1}{2}, \frac{1}{2})$.

b allele-only mixture



$\varepsilon_j \in \{\text{imbalanced, balanced}\}$; f_j: expected lesser-allele fraction of the cell's genotype (imbalanced: ϵ, $\frac{1}{3}$ or $\frac{1}{4}$; balanced: $\frac{1}{2}$); a_s: which allele of SNP s is in the minority, shared by all cells; $p_{sj} = f_j$ if $a_s = 1$, else $1 - f_j$; $r_{sj} \sim \mathrm{BetaBin}(n_{sj}, \kappa p_{sj}, \kappa(1 - p_{sj}))$.

c combined expression and allele mixture



ε_j: copy-number state (i6: 0, 1, 2, 3, 4, $\geq$5 copies and copy-neutral LOH; i3: loss, neutral, gain, copy-neutral LOH); f_j: lesser-allele fraction of one of the state's genotypes (e.g. 3 copies: $\frac{1}{3}$; LOH and 1 copy: ϵ); $\bar{x}_j$: the cell's mean residual over the region's genes, $\bar{x}_j \sim \mathcal{N}\left(\mu_{\varepsilon_j}, (s_r \sigma_{\varepsilon_j} / \sigma_{\text{neutral}})^2\right)$; allele terms as in b.